

Jaxon Cooper

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RELEVANT WORK EXPERIENCE

Airport Ramp Technician

August 2025 – Present

City of Petaluma

Petaluma, CA

- Verifies proper operation and quality of AVGAS and JET A fuel storage tanks, pumps, and fuel delivery trucks in compliance with safety procedures.
- Conducts runway inspections, aircraft fueling and marshalling, and airport grounds maintenance.

Auto Shop Assistant

November 2024 – May 2025

Cal Poly Humboldt

Arcata, CA

- Supported maintenance of university fleet vehicles and buses through cleaning, refueling, and readiness checks.
- Maintained shop organization and inventory control.

Materials Engineering Lab Assistant

December 2023 – May 2024

Santa Rosa Junior College

Santa Rosa, CA

- Assembled, calibrated, and organized materials testing equipment prior to the start of the semester.
- Troubleshooted and repaired faulty equipment throughout the semester.
- Guided students through testing equipment use during lab sessions.

PROJECTS

Capstone Project – High Precision Wood Wool Cutter

August 2025 – May 2026

Cal Poly Humboldt

Arcata, CA

- Designed and manufactured a machine that shaves wood into precisely dimensioned wood shavings for use in sustainable building materials.
- Concentration on effective blade design, precise cutting and feed rate control, and meeting strand dimension constraints.
- Rigorous experimental design includes varying blade geometry and cutting parameters for best possible wood wool quality.

Rotorcraft Service Ceiling and Time-To-Climb Predictor Algorithm

August 2025 – December 2025

Cal Poly Humboldt

Arcata, CA

- Developed a Python program which makes service ceiling and TTC predictions based on rotorcraft specifications.
- Algorithm utilizes the Dormand-Prince ODE solving method to determine threshold altitude and TTC.

VOLUNTEERING

Auxiliarist and Boat Crewmember Trainee

April 2026 – Present

U.S. Coast Guard Auxiliary

SKILLS

- **Technical Skills:** GD&T, SOLIDWORKS, CNC machining (G-code), Manual milling machines, Manual lathe machines, Metallography, LabVIEW, Oscilloscopes, Waveform generators, Ltspice, Autodesk Inventor
- **Languages:** Wolfram Mathematica, LaTeX, Python, C, MATLAB

EDUCATION

B.S. Mechanical Engineering

August 2024 – December 2026

Cal Poly Humboldt – Upper Division Transfer

Arcata, CA

- Undergraduate Research Assistant - Lee Research Group for Surgical Technology Analysis and Development
- Aviation Club Treasurer